Chapter 8. Functions Defined by Integrals

Continuity and differentiation under the integral sign, the gamma and beta functions, integral transforms

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8.1 Integrals depending on a parameter

For f on $[a,b]\times[c,d]$ let $\phi(x)=\int_c^d f(x,t)\,dt$. Write $D_1 f=\partial f/\partial x$.

Proposition 8.1 (8.1.1)

Let $f:[a,b]\times[c,d]\to\mathbb{R}$ be continuous.
(a) ϕ is continuous on $[a,b]$.
(b) If $D_1 f$ exists and is continuous on $(a,b)\times[c,d]$, then ϕ is differentiable on (a,b) and $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. (a) f is uniformly continuous on the compact rectangle (Theorem 3.18 of Volume I): given ϵ there is δ with $|f(x,t)-f(x',t)|<\epsilon$ for $|x-x'|<\delta$ and all t; then $|\phi(x)-\phi(x')|\le\epsilon(d-c)$.

(b) Fix x and a compact $[x-\eta,x+\eta]\subset(a,b)$, on which $D_1 f$ is uniformly continuous. For $0<|h|<\eta$, the mean value theorem gives $\frac{f(x+h,t)-f(x,t)}{h}=D_1 f(x+\theta h,t)$ with $0<\theta=\theta(h,t)<1$, so

$$\Big|\frac{\phi(x+h)-\phi(x)}{h}-\int_c^d D_1 f(x,t)\,dt\Big|\le\int_c^d |D_1 f(x+\theta h,t)-D_1 f(x,t)|\,dt\le\epsilon(d-c)$$

once $|h|$ is small. ∎

Example 8.1

Differentiating $\int_0^1 t^x\,dt=\frac{1}{x+1}$ n times in $x>-1$ gives $\int_0^1 t^x(\log t)^n dt=(-1)^n n!/(x+1)^{n+1}$. Interchanging the order, $\int_a^b\phi(x)\,dx=\int_c^d\int_a^b f(x,t)\,dx\,dt$ for continuous f: both sides, as functions of the upper limit b, vanish at $b=a$ and have the same derivative $\int_c^d f(b,t)\,dt$ (by (a), (b) and Theorem 5.11 of Volume I).

For f continuous on $[a,b]\times[c,d)$ $(d\le\infty)$, consider the improper integral $\phi(x)=\int_c^d f(x,t)\,dt$.

Theorem 8.2 (8.1.2)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, and suppose there is $g:[c,d)\to\mathbb{R}$ with (a) $|f(x,t)|\le g(t)$ for all (x,t) and (b) $\int_c^d g<\infty$. Then $\int_c^d f(x,t)\,dt$ converges for each x and ϕ is continuous on $[a,b]$.

Proof. Convergence follows from the comparison test (Proposition 7.8). Given ϵ, choose $d'<d$ with $\int_{d'}^d g<\epsilon$. Then $|\phi(x)-\phi_{d'}(x)|<\epsilon$ for all x, where $\phi_{d'}(x)=\int_c^{d'} f(x,t)\,dt$ is continuous by Proposition 8.1. So ϕ is a uniform limit of continuous functions (Theorem 6.1). ∎

Theorem 8.3 (8.1.3)

Let $f:[a,b]\times[c,d)\to\mathbb{R}$ be continuous, $D_1 f$ exist and be continuous on $(a,b)\times[c,d)$, and $\int_c^d f(x_0,t)\,dt$ converge for some $x_0\in(a,b)$. If there is $h:[c,d)\to\mathbb{R}$ with (a) $|D_1 f(x,t)|\le h(t)$ and (b) $\int_c^d h<\infty$, then $\int_c^d f(x,t)\,dt$ converges for each $x\in(a,b)$, and ϕ is differentiable with $\phi'(x)=\int_c^d D_1 f(x,t)\,dt$.

Proof. By the mean value theorem, $|f(x,t)-f(x_0,t)|\le|x-x_0|h(t)$, so $\int_c^d f(x,t)dt$ converges by comparison. For $c<d'<d$ let $\phi_{d'}(x)=\int_c^{d'} f(x,t)dt$; then $\phi'_{d'}(x)=\int_c^{d'} D_1 f(x,t)dt$ (Proposition 8.1), and these converge uniformly in x as $d'\to d$, because the tails are bounded by $\int_{d'}^d h$. Also $\phi_{d'}(x_0)\to\phi(x_0)$. Applying Theorem 6.4 along any sequence $d'_n\to d$ gives the result. ∎

Example 8.2

(a) $F(x)=\int_0^\infty e^{-t^2}\cos(xt)\,dt$ satisfies $F'(x)=-\int_0^\infty te^{-t^2}\sin(xt)\,dt=-\frac{x}{2}F(x)$ (integration by parts), so $F(x)=F(0)e^{-x^2/4}=\frac{\sqrt{\pi}}{2}e^{-x^2/4}$.
(b) $\int_0^\infty \frac{\sin x}{x}dx=\frac{\pi}{2}$: for $G(s)=\int_0^\infty e^{-sx}\frac{\sin x}{x}dx$ $(s>0)$, Theorem 8.3 gives $G'(s)=-\frac{1}{1+s^2}$, and $G(s)\to 0$ as $s\to\infty$, so $G(s)=\frac{\pi}{2}-\arctan s$; continuity at $s=0$ follows from the second mean value theorem (Corollary 5.25 of Volume I).

8.2 The gamma function

For $x>0$ let $\Gamma(x)=\int_0^\infty t^{x-1}e^{-t}\,dt$, convergent since $t^{x-1}e^{-t}\le t^{x-1}$ near 0 and $\le Ct^{-2}$ near ∞. By Theorem 8.3 on compact subintervals of $(0,\infty)$, Γ is C^∞ with $\Gamma^{(n)}(x)=\int_0^\infty t^{x-1}(\log t)^n e^{-t}dt$. Integration by parts gives $\Gamma(x+1)=x\Gamma(x)$, and $\Gamma(1)=1$, so $\Gamma(n+1)=n!$.

A function $f:I\to\mathbb{R}$ is **convex** if $f((1-\lambda)x+\lambda y)\le(1-\lambda)f(x)+\lambda f(y)$ for $x,y\in I$, $0\le\lambda\le 1$.

Proposition 8.4 (8.2.1)

(a) A continuous $f:I\to\mathbb{R}$ with $f(\frac{x+y}{2})\le\frac{f(x)+f(y)}{2}$ for all $x,y\in I$ is convex.
(b) A convex f satisfies, for $s<t<u$ in I,

$$\frac{f(t)-f(s)}{t-s}\le\frac{f(u)-f(s)}{u-s}\le\frac{f(u)-f(t)}{u-t}.$$

Proof. (a) By induction on n, the convexity inequality holds for dyadic $\lambda=k/2^n$: if $\lambda=\frac{\lambda_1+\lambda_2}{2}$ with dyadic λ_i of level $n-1$, then $(1-\lambda)x+\lambda y$ is the midpoint of the points for λ_1,λ_2. For general λ, approximate by dyadic $\lambda_n\to\lambda$ and use continuity.

(b) Write $t=(1-\lambda)s+\lambda u$ with $\lambda=\frac{t-s}{u-s}$; then $f(t)\le(1-\lambda)f(s)+\lambda f(u)$, which rearranges to each of the two inequalities. ∎

$\log\Gamma$ is convex: by Hölder's inequality with exponents $1/\lambda$, $1/(1-\lambda)$ (or Cauchy–Schwarz for the midpoint and Proposition 8.4(a)),

$$\Gamma\Big(\frac{x+y}{2}\Big)=\int_0^\infty t^{\frac{x-1}{2}}e^{-t/2}\,t^{\frac{y-1}{2}}e^{-t/2}\,dt\le\Gamma(x)^{1/2}\Gamma(y)^{1/2}.$$

Theorem 8.5 (8.2.2, Bohr–Mollerup)

If $f:(0,\infty)\to(0,\infty)$ satisfies (a) $f(x+1)=xf(x)$, (b) $f(1)=1$, (c) $\log f$ is convex, then $f=\Gamma$.

Proof. Γ has these properties, so it suffices to show f is determined by them; by (a) it suffices to determine f on $(0,1]$. Let $\phi=\log f$, $0<x\le 1$, $n\in\mathbb{N}$. By (a), (b), $\phi(n+1)=\log n!$. Apply Proposition 8.4(b) to the points $n<n+1<n+1+x$ and $n+1<n+1+x<n+2$:

$$\log n\le\frac{\phi(n+1+x)-\phi(n+1)}{x}\le\log(n+1).$$

By (a), $\phi(n+1+x)=\phi(x)+\log[x(x+1)\cdots(x+n)]$. Hence

$$0\le\phi(x)-\log\frac{n!\,n^x}{x(x+1)\cdots(x+n)}\le x\log\Big(1+\frac{1}{n}\Big)\to 0,$$

so $f(x)=\lim_n\frac{n!\,n^x}{x(x+1)\cdots(x+n)}$ (Gauss's formula) is determined. ∎

Example 8.3

The **beta function** $B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ satisfies $B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$: for fixed y, $f(x) = \Gamma(x+y)B(x,y)/\Gamma(y)$ satisfies (a), (b), (c) of Theorem 8.5. In particular $\Gamma(\frac{1}{2})^2 = B(\frac{1}{2},\frac{1}{2}) = \pi$, so $\int_{-\infty}^{\infty} e^{-x^2}dx = \Gamma(\frac{1}{2}) = \sqrt{\pi}$. Stirling's formula $\Gamma(x+1) \sim \sqrt{2\pi x}\, x^x e^{-x}$ follows from the substitution $t = x(1+u)$ and Laplace's method.